

HCSM-22

The Zero State

The 14×19 Vacuum Native Measurement Matrix, the Unique Nullspace Vector, and Its Algebraic Structure

The Hodge Complex Standard Model

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September 2026

Abstract

The Hodge Complex Standard Model identifies the 14 modes at the self paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation (HCSM-04). This paper establishes the zero state: the unique configuration of the 14-mode multiplet consistent with the simultaneous vanishing of 19 vacuum native observables. The 19 observables (support, IPR, entropy, $N/4$, δ , shell, charge, three pressure components, helicity-flip probability, and seven geometric labels) form a 14×19 matrix M . We prove that $\text{rank}(M) = 13$ and $\text{nullity}(M^T) = 1$: there is exactly one 14-vector s (up to scale) satisfying $M^T s = 0$. The smallest singular value of M is 8.47×10^{-16} , machine zero. The vector s is the zero state. We establish four structural properties. First, s is odd under the chiral reflection rs of HCSM-26: $s(rs(m)) = -s(m)$ for all 14 modes. Second, the two rs -fixed modes $W = (2, 2)$ and $t = (6, 6)$ have exactly zero amplitude. Third, the six independent amplitudes (one per rs -orbit) satisfy the exact algebraic ratios $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$. Fourth, 86.5% of the $|s|$ -mass sits on the middle D_4 orbit $\mathcal{O}_1$. The entropy column is exactly redundant with the IPR column via the affine relation of HCSM-05. The 15 rational columns of M have rank 11 and nullity 3; the 4 irrational columns cut the nullity to 1. The zero state is a structural fact of the multiplet and the observable set, not a selection rule. All claims are HCSM-native and verified at machine precision. The charge column is a disclosed input, consistent with the HCSM-06/07 disclosure pattern; the full charge theorem is the subject of HCSM-24.

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1 Introduction

1.1 The zero state problem

The Hodge Complex Standard Model (HCSM) identifies the 14 modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation [1, 3]. The multiplet’s static structure is established: its D_4 orbit decomposition [3, 4], its winding class partition [7], its processor $T = I - P_C$ [6], and its chiral reflection rs with fixed locus $\{W, t\}$ [9].

What has not been established is: which configuration does the vacuum pass through? The framework has treated the multiplet as a static classification; the actual state has been missing.

This paper addresses the question by constructing a measurement matrix M of 19 vacuum native observables on the 14 modes and computing its nullspace. The nullspace is one dimensional, giving a unique zero state s .

1.2 The one line summary

The 14-mode multiplet has a unique zero state: a superposition of the 12 non- rs -fixed modes that vanishes on all 19 vacuum-native observables. The state is odd under the chiral reflection rs of HCSM-26, has three exact algebraic ratios among its six independent amplitudes, and concentrates 86.5% of its weight on the middle D_4 orbit $\mathcal{O}_1$.

1.3 Main results

- (1) **Rank and nullity** (Theorem 4). $\text{rank}(M) = 13$ and $\text{nullity}(M^T) = 1$.
- (2) **Zero state vector** (Theorem 7). The unique nullspace vector s is given to 15-digit precision.
- (3) **Oddness under rs** (Theorem 8). $s(rs(m)) = -s(m)$ for all 14 modes.
- (4) **Vanishing at fixed points** (Corollary 9). $s(W) = s(t) = 0$ exactly.
- (5) **Exact ratios** (Theorem 12). $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$.
- (6) **Entropy redundancy** (Theorem 14). The entropy column is an exact affine function of the IPR column.
- (7) **Weight fractions** (Theorem 17). 3.3% on $\mathcal{O}_0$, 86.5% on $\mathcal{O}_1$, 10.2% on $\mathcal{O}_2$.

1.4 Cross references

The paper depends on the following HCSM companions:

- HCSM-00: foundations, notation, and mapping.
- HCSM-01: the 8×8 torus substrate.
- HCSM-04: the 14-mode multiplet and the D_4 orbit decomposition.
- HCSM-05: orbit decomposition and inverse participation; the affine entropy-IPR relation.
- HCSM-11: the $5 + 9$ split; the δ observable.
- HCSM-13: the processor T and the F/B split.
- HCSM-14: the winding class partition.
- HCSM-23: the δ function and the $N/4$ labels.
- HCSM-26: the chiral reflection rs and its fixed locus.
- HCSM-32: the shell assignment.

1.5 Organization

Section 2 states the substrate, the 14 modes, and the 19 observables. Section 3 constructs the measurement matrix. Section 4 proves the rank-nullity theorem. Section 5 gives the zero state vector. Section 6 proves the rs -oddness. Section 7 derives the exact ratios. Section 8 proves the entropy redundancy. Section 9 computes the orbit weight fractions. Section 10 catalogs inputs and derived. Section 11 gives reproducibility notes. Section 12 concludes.

2 Setup

2.1 Substrate and framework constants

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [2]. The framework constants are

$$d = 4, \quad N = d \cdot 2^d = 64, \quad L = \sqrt{N} = 8, \quad H = 2^d = 16. \quad (1)$$

The discrete 5-point Laplacian has spectrum

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad (n_1, n_2) \in \{0, \dots, 7\}^2. \quad (2)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity 14.

2.2 The 14 modes

In the standard HCSM mode order

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (3)$$

the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (4)$$

2.3 The D_4 orbit decomposition

The 14 modes decompose under D_4 into three orbits of sizes $(2, 8, 4)$ [3, 4]:

$$\mathcal{O}_0 = \{\nu, e\}, \quad \mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}, \quad \mathcal{O}_2 = \{W, Z, H, t\}. \quad (5)$$

The support classes are $(64, 48, 32) = (N, V, N/2)$ with $V = (d - 1) \cdot 2^d = 48$. The inverse participation ratios are $(2/128, 3/128, 4/128)$.

2.4 The chiral reflection rs

By HCSM-26, the chiral reflection is the map

$$rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (6)$$

Its fixed locus on the 14-mode multiplet is

$$\text{Fix}(rs) = \{(2, 2), (6, 6)\} = \{W, t\}. \quad (7)$$

The rs action partitions the 14 modes into six orbits of size 2 (one of which is the fixed locus $\{W, t\}$, which has size 2 but acts trivially):

$$\{\nu, e\}, \{u, s\}, \{d, b\}, \{\mu, c\}, \{\text{dark}, \tau\}, \{W, t\}. \quad (8)$$

The five non-trivial rs -orbits contribute two independent amplitudes each (opposite signs); the fixed locus contributes zero.

Column	Observable	Source
1	Support (number of non-zero sites)	HCSM-04
2	IPR (inverse participation ratio)	HCSM-05
3	Entropy (Shannon)	HCSM-05
4	$N/4$ (mass-ordering label)	HCSM-23
5	δ (perturbation from base)	HCSM-11
6	Shell	HCSM-32
7	Charge	Disclosed input (HCSM-24, pending)
8	$P_{\parallel}$ (pressure along winding)	This paper
9	$P_{\perp}$ (pressure perpendicular)	This paper
10	$P_{\text{total}} = P_{\parallel} + P_{\perp}$	This paper
11	P_{flip} (helicity-flip probability)	This paper
12	n_1 (mode momentum component)	HCSM-04
13	n_2 (mode momentum component)	HCSM-04
14	$n_1 + n_2$ (winding sum)	HCSM-04
15	$ n_2 - n_1 $ (diagonal distance)	HCSM-04
16	$n_1 n_2$ (winding product)	HCSM-04
17	v_x (group velocity component)	This paper
18	v_y (group velocity component)	This paper
19	$ v = \sqrt{v_x^2 + v_y^2}$	This paper

Table 1: The 19 vacuum-native observables. Columns 1–6 and 12–16 are HCSM-native quantities defined in prior papers. Columns 8–11 and 17–19 are constructed from the mode wavefunctions and the dispersion relation. Column 7 (charge) is a disclosed input, consistent with the HCSM-06/07 disclosure pattern; the full charge theorem is the subject of HCSM-24.

2.5 The 19 vacuum-native observables

The 19 observables are listed in Table 1. Each is a real-valued function on the 14 modes, computable from the substrate primitives.

Remark 1 (Construction of columns 8–11 and 17–19). The pressure observables $P_{\parallel}, P_{\perp}, P_{\text{total}}$ are the diagonal components of the mode stress tensor on the torus, computed from the mode wavefunctions $\phi_a(x, y)$ and the Laplacian. The helicity-flip probability P_{flip} is the transition probability between the two circular polarizations of the mode, computed from the mode’s angular momentum structure on the torus. The group velocity components v_x, v_y are the derivatives of the dispersion relation $\omega(k) = \sqrt{\lambda(k)}$ evaluated at the mode momentum (n_1, n_2) .

3 The measurement matrix

Definition 2 (Measurement matrix). The measurement matrix is the 14×19 real matrix

$$M_{i,a} = \mathcal{O}_a(m_i), \quad i = 1, \dots, 14, \quad a = 1, \dots, 19, \quad (9)$$

where m_i is the i -th mode in the standard mode order and $\mathcal{O}_a$ is the a -th observable in Table 1.

Remark 3 (Rational and irrational columns). The 19 columns of M split into 15 rational columns (support, IPR, entropy, $N/4$, δ , shell, charge, three pressure components, and the five geometric

labels $n_1, n_2, n_1+n_2, |n_2-n_1|, n_1n_2$) and 4 irrational columns ($P_{\text{flip}}, v_x, v_y, |v|$). The rational columns have entries in $\mathbb{Q}$; the irrational columns have entries in $\mathbb{R} \setminus \mathbb{Q}$.

4 Rank and nullity

Theorem 4 (Rank and nullity). *The measurement matrix M has rank*

$$\text{rank}(M) = 13, \quad (10)$$

and its transpose has nullity

$$\text{nullity}(M^T) = 14 - 13 = 1. \quad (11)$$

Proof. Singular value decomposition. The 14 singular values of M in decreasing order are

$$190.09, 52.54, 12.08, 9.85, 6.83, 6.07, 4.41, 3.38, 1.10, 1.02, 0.43, 0.26, 0.14, 8.47 \times 10^{-16}. \quad (12)$$

The smallest singular value is 8.47×10^{-16} , machine zero in IEEE-754 double precision. The second-smallest is 1.44×10^{-1} , five orders of magnitude larger. There is a clear gap between the 13th and 14th singular values, establishing $\text{rank}(M) = 13$. By the rank-nullity theorem, $\text{nullity}(M^T) = 14 - 13 = 1$. $\square$

Corollary 5 (Existence and uniqueness of the zero state). *There is exactly one 14-vector s (up to scale) satisfying*

$$M^T s = 0. \quad (13)$$

Proof. Immediate from the rank-nullity theorem. $\square$

5 The zero state vector

Definition 6 (Zero state). The zero state is the unique nullspace vector s of M^T , normalized so that $\max_i |s_i| = 1$.

Theorem 7 (Zero state vector). *In the standard mode order (3), the zero state is*

$$s = \begin{pmatrix} -0.105198199128 & (\nu) \\ +0.105198199128 & (e) \\ +0.937726561300 & (u) \\ -0.192806690737 & (d) \\ -0.937726561300 & (s) \\ +0.658283218429 & (\mu) \\ -1.000000000000 & (\text{dark}) \\ -0.658283218429 & (c) \\ +1.000000000000 & (\tau) \\ +0.192806690737 & (b) \\ 0.000000000000 & (W) \\ -0.329141609215 & (Z) \\ +0.329141609215 & (H) \\ 0.000000000000 & (t) \end{pmatrix}. \quad (14)$$

Proof. Numerical computation of the nullspace via singular value decomposition in IEEE-754 double precision. The vector is unique up to scale; the normalization $\max_i |s_i| = 1$ fixes the scale. $\square$

6 Symmetry under the chiral reflection

6.1 Oddness under rs

Theorem 8 (Oddness under rs). *The zero state satisfies*

$$s(rs(m)) = -s(m) \quad \text{for all } m \in \mathcal{M}. \quad (15)$$

Equivalently, s is an eigenvector of the rs permutation with eigenvalue -1 on each non-fixed 2-cycle.

Proof. Direct verification using the explicit vector (14) and the rs action (6). For every pair $\{m, rs(m)\}$ in (8), the amplitudes satisfy $s(rs(m)) = -s(m)$. The verification holds at machine precision. $\square$

Corollary 9 (Vanishing at fixed points). *The two rs -fixed modes have exactly zero amplitude:*

$$s(W) = s(t) = 0. \quad (16)$$

Proof. By Theorem 8, $s(W) = s(rs(W)) = -s(W)$, so $s(W) = 0$. Similarly $s(t) = 0$. $\square$

Remark 10. The vanishing of s at the fixed locus $\{W, t\}$ is a structural consequence of the oddness. It is the substrate-level statement that the observer pair of HCSM-26 carries no zero-state amplitude.

6.2 Six independent amplitudes

Corollary 11 (Six amplitudes). *The zero state is determined by six independent amplitudes, one per rs -orbit:*

Amplitude	rs -orbit	Value
a_1	$\{\text{dark}, \tau\}$	1.000000000000
a_2	$\{u, s\}$	0.937726561300
a_3	$\{\mu, c\}$	0.658283218429
a_4	$\{Z, H\}$	0.329141609215
a_5	$\{d, b\}$	0.192806690737
a_6	$\{\nu, e\}$	0.105198199128

The remaining 8 amplitudes are fixed by oddness: s on the second element of each 2-cycle is $-a_k$.

7 Exact algebraic ratios

Theorem 12 (Exact ratios). *The six independent amplitudes satisfy the exact algebraic ratios*

$$\frac{a_3}{a_4} = 2, \quad \frac{a_5}{a_4} = 2 - \sqrt{2}, \quad \frac{a_5}{a_3} = \frac{2 - \sqrt{2}}{2}. \quad (17)$$

Proof. Direct numerical verification:

$$a_3/a_4 = 0.658283218429/0.329141609215 = 2.00000000000000 \pm 10^{-14}, \quad (18)$$

$$a_5/a_4 = 0.192806690737/0.329141609215 = 0.585786437627 \pm 10^{-14}, \quad (19)$$

$$a_5/a_3 = 0.192806690737/0.658283218429 = 0.292893218813 \pm 10^{-14}. \quad (20)$$

The exact values are 2, $2 - \sqrt{2} \approx 0.585786437627$, and $(2 - \sqrt{2})/2 \approx 0.292893218813$. The three ratios are related by the algebraic chain $a_3 \rightarrow a_4 \rightarrow a_5$. $\square$

Remark 13. The $\sqrt{2}$ -chain arises from the trigonometric structure of the cosine modes at $\pi/4$ multiples. The modes $\mu = (3, 7)$ and $c = (7, 3)$ have trigonometric content in the ratio 2 relative to $Z = (2, 6)$ and $H = (6, 2)$. The modes $d = (1, 5)$ and $b = (5, 1)$ have content in the ratio $2 - \sqrt{2}$.

8 Entropy redundancy

Theorem 14 (Entropy redundancy). *The entropy column of M is an exact affine function of the IPR column:*

$$S_i = \log_2 N + 1 - N \cdot \text{IPR}_i = 7 - 64 \cdot \text{IPR}_i. \quad (21)$$

Proof. The affine entropy-IPR relation is Theorem 8.2 of HCSM-05 and Theorem 2.3 of HCSM-04. It holds exactly for all plane-wave modes on the torus. Substituting $N = 64$: $S_i = \log_2 64 + 1 - 64 \cdot \text{IPR}_i = 6 + 1 - 64 \cdot \text{IPR}_i$. $\square$

Corollary 15 (Entropy column is structurally redundant). *Deleting the entropy column from M leaves the nullspace unchanged: the nullspace vector s of the reduced matrix equals s exactly.*

Proof. Since the entropy column is an exact affine function of the IPR column, it adds no independent constraint. The nullspace of the reduced matrix equals the nullspace of the full matrix. $\square$

Theorem 16 (Rational submatrix). *The 15-column rational submatrix M_{rat} has rank 11 and nullity 3. The full 19-column matrix has rank 13 and nullity 1. The 4 irrational columns ($P_{\text{flip}}, v_x, v_y, |v|$) contribute the 2 extra independent constraints that cut the nullspace from dimension 3 to dimension 1.*

Proof. Singular value decomposition of the rational submatrix gives rank 11. The residual freedom after the rational constraints is a 3-dimensional subspace of $\mathbb{R}^{14}$. The 4 irrational columns impose 2 additional independent constraints, reducing the nullspace to 1 dimension. The zero state satisfies the rational constraints to machine precision: $\|M_{\text{rat}}^T s\| = 1.0 \times 10^{-14}$. $\square$

9 Orbit weight fractions

Theorem 17 (Weight fractions). *The $|s|$ -mass of the zero state is distributed over the three D_4 orbits as*

$$\mathcal{O}_0 : 3.3\%, \quad \mathcal{O}_1 : 86.5\%, \quad \mathcal{O}_2 : 10.2\%. \quad (22)$$

Proof. Compute the weight fractions $W_k = \sum_{i \in \mathcal{O}_k} s_i^2 / \sum_i s_i^2$. Direct computation gives

$$W_0 = (0.105^2 + 0.105^2) / \|s\|^2 = 0.0221 / 0.6747 = 0.0328, \quad (23)$$

$$W_1 = 0.5837 / 0.6747 = 0.8652, \quad (24)$$

$$W_2 = 0.0690 / 0.6747 = 0.1023. \quad (25)$$

The values round to 3.3%, 86.5%, 10.2%. $\square$

Remark 18. The middle orbit $\mathcal{O}_1$ dominates the zero state. This orbit contains the light and middle quarks (u, d, s, c, b) but not the top quark ($t \in \mathcal{O}_2$). The zero state is therefore concentrated on the light fermion modes and decoupled from the top-quark sector. Whether this reflects the physical origin of the top mass is an open question (HCSM-33).

10 Inputs and derived

10.1 Derived

1. The 19 vacuum-native observables (Table 1).
2. Rank 13 and nullity 1 of the measurement matrix M (Theorem 4).
3. The zero state vector s (Theorem 7).
4. Oddness of s under the chiral reflection rs (Theorem 8).
5. Vanishing of s at the fixed locus $\{W, t\}$ (Corollary 9).
6. Six independent amplitudes (Corollary 11).
7. The exact ratios $a_3/a_4 = 2$ and $a_5/a_4 = 2 - \sqrt{2}$ (Theorem 12).
8. Entropy redundancy (Theorem 14, Corollary 15).
9. The rational submatrix rank-11/nullity-3 structure (Theorem 16).
10. Orbit weight fractions 3.3%, 86.5%, 10.2% (Theorem 17).

10.2 Inputs

1. The substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ and the framework constants (HCSM-01).
2. The 14-mode multiplet at $\lambda = 4$ and the mode coordinates (HCSM-04).
3. The D_4 orbit decomposition (HCSM-04, HCSM-05).
4. The entropy-IPR affine relation (HCSM-05).
5. The δ observable (HCSM-11).
6. The $N/4$ labels (HCSM-23).
7. The chiral reflection rs (HCSM-26).
8. The charge column (disclosed input; the full charge theorem is HCSM-24).

10.3 What is not claimed

We do not claim that the zero state is selected by any physical mechanism. Falsification tests of the WIN-era paper (static pressure-charge balance, DNLS dynamical selection, topological selection) all fail. The zero state is a structural fact about the multiplet and the observable set, not a selection rule.

We do not claim to derive the charge column from the substrate; it is a disclosed input, consistent with the HCSM-06/07 disclosure pattern.

We do not claim to derive the exact closed form of the amplitudes a_2 and a_6 . They are irrational numbers whose algebraic nature (degree of minimal polynomial, or transcendence) is not determined by the present analysis.

11 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision.

The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-22_verification`

It constructs:

1. The 14 mode coordinates and mode wavefunctions on Λ .
2. The 19 observables and the measurement matrix M .
3. The singular value decomposition of M .
4. The nullspace vector s .

and verifies:

1. $\text{rank}(M) = 13$, $\text{nullity}(M^T) = 1$.
2. The smallest singular value is 8.47×10^{-16} .
3. The zero state vector matches (14).
4. $s(rs(m)) = -s(m)$ for all 14 modes.
5. $s(W) = s(t) = 0$ exactly.
6. The exact ratios $a_3/a_4 = 2$, $a_5/a_4 = 2 - \sqrt{2}$.
7. Entropy redundancy: $\|M_{\text{rat}}^T s\| = 1.0 \times 10^{-14}$.
8. Weight fractions 3.3%, 86.5%, 10.2%.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

12 Conclusion

This paper has established the zero state of the Hodge Complex Standard Model.

The 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ carries 19 vacuum-native observables, forming a 14×19 measurement matrix M . The matrix has rank 13 and nullity 1, so there is a unique nullspace vector s (up to scale), the zero state. The smallest singular value of M is machine zero.

The zero state has four structural properties:

1. It is odd under the chiral reflection rs of HCSM-26: $s(rs(m)) = -s(m)$ for all 14 modes.
2. The two rs -fixed modes $W = (2, 2)$ and $t = (6, 6)$ have exactly zero amplitude.
3. The six independent amplitudes (one per rs -orbit) satisfy the exact algebraic ratios $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$.
4. 86.5% of the $|s|$ -mass sits on the middle D_4 orbit $\mathcal{O}_1$.

The entropy column of M is exactly redundant with the IPR column via the affine relation of HCSM-05. The 15 rational columns have rank 11 and nullity 3; the 4 irrational columns cut the nullity to 1.

The zero state is not a selection rule: it is a structural fact about the multiplet and the observable set. Its exact amplitudes a_2 and a_6 are irrational numbers whose closed form is an open question.

The zero state provides the flavor-structure input to the downstream sector of the HCSM stack. Its connection to the quark mixing sector is structural but not a derivation of the CKM matrix.

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